Preprint

A Nonlocal, Baryon-Anchored Analysis of Galactic Acceleration Excess

Ukshin Q. Rexhepi

Independent Researcher
Tübingen, Germany

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Abstract. This study presents a fully nested, galaxy-separated test of whether the positive galactic acceleration excess in SPARC rotation curves can be reconstructed from a nonlocal, scale-dependent, baryon-anchored model family. The analysis uses 175 SPARC galaxies and 3391 radial measurements. The nonlocal model reaches prediction accuracy statistically indistinguishable from a foldwise fitted radial acceleration relation (RAR). The dynamical kernel remains positive in all 4500 galaxy-wise bootstrap fits, and a leave-one-out kernel with its local self-contribution removed remains positive in all outer-fold models. A conditional out-of-fold galaxy-profile reassignment test yields $p_{\mathrm{MC}} = 1/2001$ in all three quality samples. An additional soft-saturation correction shows no consistent incremental predictive gain and does not determine a unique transition scale. A fixed nonlocal $q = 0.5$ gradient response is therefore specified only as a parsimonious candidate for future external testing. The results identify a reproducible cross-radial, baryon-anchored, scale-dependent predictive structure in the SPARC development sample. They do not establish a fundamental physical mechanism, exclude particle dark matter, or constitute external validation of the frozen candidate model.

Keywords: dark matter phenomenology; galactic rotation curves; SPARC; radial acceleration relation; nonlocal empirical model; scale dependence; nested galaxy-separated cross-validation; external validation

1. Introduction

1.1. Aim of the Empirical Analysis

This work investigates galactic dark-matter phenomenology through a reproducible field-response analysis with strict galaxy separation. The observed additional acceleration component is not assumed at the outset to result from a separate invisible particle substance. It is first treated as an empirical target quantity whose dependence on baryonic organization, radial structure, and global galactic scale can be tested directly.

This does not deny the dark-matter effect. The positive acceleration excess

$$g_{\mathrm{field,obs}}(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r) > 0$$

is used as a real dynamical observable. The central question is whether this excess can be reconstructed in wholly held-out galaxies from a nonlocal measure of baryonic organization, normalized radial position, local disk structure, and galaxy-wide scale features.

The analysis pursues two separate aims. First, an empirical model family is tested under fully nested GroupKFold cross-validation. Kernel scale, global feature form, and regularization are determined solely within the training galaxies. The empirical RAR serves as a strong reference relation and is refitted fold by fold under the same outer galaxy separation. Second, the analysis determines which elements of the field-response structure are statistically stable and which can only be regarded as secondary hypotheses.

The analysis is deliberately model-agnostic. It tests a concrete minimum empirical structure through comparison with the RAR, a conditional out-of-fold galaxy-profile reassignment test, galaxy-wise coefficient bootstraps, and generic machine-learning baselines. It neither derives a fundamental field equation nor attributes the observed structure to a unique physical mechanism.

MDAR and BTFR are treated as secondary consistency analyses. They do not replace independent rotation-curve data with fully propagated observational and mass-model uncertainties.

2. Materials and Methods

2.1. Data Basis and Target Quantity

The analysis uses the SPARC rotation curves and mass models of 175 disk galaxies [1]. A total of 3391 radial measurements were processed. The observed rotation velocities and the baryonic contributions of gas, disk, and, where applicable, bulge were used to calculate

$$g_{\mathrm{obs}}(r) = \frac{V_{\mathrm{obs}}^2(r)}{r}$$

and

$$g_{\mathrm{bar}}(r) = \frac{V_{\mathrm{gas}}(r)|V_{\mathrm{gas}}(r)| + \Upsilon_{\mathrm{disk}} V_{\mathrm{disk}}^2(r) + \Upsilon_{\mathrm{bulge}} V_{\mathrm{bulge}}^2(r)}{r}$$

in SI units. The adopted values are $\Upsilon_{\mathrm{disk}} = 0.5$ and $\Upsilon_{\mathrm{bulge}} = 0.7$. The gas contribution retains its sign, following the convention of the SPARC Rotmod files.

The regression target is the dimensionless logarithm

$$y(r) = \log_{10}\left(\frac{g_{\mathrm{field,obs}}(r)}{g_0}\right), \qquad g_{\mathrm{field,obs}} = g_{\mathrm{obs}} - g_{\mathrm{bar}} > 0,$$

with the reference acceleration $g_0 = 1\,\mathrm{m\,s^{-2}}$. This notation is numerically identical to the implementation in fixed SI units, while making the logarithm dimensionless. The subscript

"field" is retained as a model label for the reconstructed excess and does not imply a specific field ontology or physical mechanism. After validity, positivity, and complete-feature filters, the analysed samples comprise:

Table 1: Positive complete analysis points and galaxies for each quality selection.

Sample	Radial points	Galaxies
$Q = 1$	1859	99
$Q = 1 + 2$	2805	163
$Q = 1 + 2 + 3$	2896	175

2.2. Positivity and Uncertainty Sensitivity

The primary logarithmic model requires $g_{\mathrm{field,obs}} > 0$. To determine whether its result is driven by formally positive points that are nevertheless close to zero relative to the observed velocity error, three stricter filters were evaluated from the beginning. The SPARC velocity uncertainty $\sigma_{V,\mathrm{obs}}$ was first used to calculate the acceleration uncertainty attributable to the observed velocity error alone,

$$\sigma_{g,\mathrm{obs}}^{(V)} = \frac{2V_{\mathrm{obs}}\sigma_{V,\mathrm{obs}}}{r}$$

in the same units as g_{obs}. The dimensionless excess indicator is

$$Z_{\mathrm{obs}} = \frac{g_{\mathrm{field,obs}}}{\sigma_{g,\mathrm{obs}}^{(V)}}.$$

Alongside the primary selection $g_{\mathrm{field,obs}} > 0$, thresholds of $Z_{\mathrm{obs}} > 0.5$, $Z_{\mathrm{obs}} > 1$, and $Z_{\mathrm{obs}} > 2$ were tested. For every threshold, kernel scale, global feature form, and ridge regularization were reselected solely within the outer training galaxies. The indicator deliberately includes only the propagated error of V_{obs}; it therefore does not fully capture uncertainties in baryonic decomposition, distance, and inclination.

2.3. Complete Definition of the Log-Radial Coupling Kernel

For a galaxy with N_g strictly positive observed radii r_i, the non-negative dynamical baryonic source quantity at radial point r_j is defined as

$$W_{\mathrm{dyn},j} = |V_{\mathrm{gas},j}|^2 + \Upsilon_{\mathrm{disk}} V_{\mathrm{disk},j}^2 + \Upsilon_{\mathrm{bulge}} V_{\mathrm{bulge},j}^2$$

Non-finite or non-positive source values are set to zero before averaging. The gas contribution deliberately differs here from the signed SPARC convention in g_{bar}: the kernel uses the positive strength of the dynamical source contribution.

The coupling of two measured radii is defined in logarithmic radius space:

$$K_{ij}^{(\lambda)} = \exp\left[-\frac{|\ln r_i - \ln r_j|}{\lambda}\right] = \exp\left[-\frac{|\ln(r_i/r_j)|}{\lambda}\right],$$

where $\lambda > 0$ is dimensionless. The kernel value used in the main model is the normalized

weighted mean

$$F_{\mathrm{dyn},\lambda}(r_i) = \frac{\sum_{j=1}^{N_g} K_{ij}^{(\lambda)} W_{\mathrm{dyn},j}}{\sum_{j=1}^{N_g} K_{ij}^{(\lambda)}}.$$

It has the same unit as W_{dyn}, namely $(\mathrm{km\,s^{-1}})^2$. The regression feature is written in dimensionless form as

$$x_{\mathrm{dyn},\lambda}(r_i) = \log_{10}\left[\frac{F_{\mathrm{dyn},\lambda}(r_i)}{F_0}\right], \qquad F_0 = 1\,(\mathrm{km\,s^{-1}})^2.$$

The main kernel includes the self-contribution $K_{ii} = 1$. No weighting by Δr_j or $\Delta \ln r_j$ is used; the sum is therefore a discrete kernel-weighted point average rather than a quadrature of a continuous integral. The sums run solely over the actually observed radial points of the respective galaxy. Neither extrapolation nor reflection is used beyond the smallest and largest measurement radius; at each radius, the normalization is recomputed over the available support, truncated accordingly.

Additional robustness controls test a leave-one-out kernel with $K_{ii} = 0$, a variant weighted by local cell widths in $\Delta \ln r$, and a sign-preserving gas-source variant. The leave-one-out variant contains contributions from other radial points only and thereby isolates the cross-radial information component.

2.4. Nonlocal Scale-Dependent Model Family

The primary model describes the dimensionless logarithmic positive acceleration excess through the coupling kernel, normalized radial location, a local disk-structure quantity, and optional galaxy-wide scale features:

$$y = a + \alpha x_{\mathrm{dyn},\lambda} + \beta R + \gamma D + \sum_j \delta_j G_j.$$

Here,

$$R = \log_{10}(r/R_d)$$

already dimensionless. Operationally, the local disk-structure quantity is the disk surface-brightness value tabulated in the SPARC Rotmod files at the measurement radius,

$$\Sigma_{\mathrm{disk,local}}(r_i) = SB_{\mathrm{disk}}(r_i),$$

and is used as

$$D = \log_{10}\left(\frac{\Sigma_{\mathrm{disk,local}}}{\Sigma_0}\right), \qquad \Sigma_0 = 1\,L_\odot\,\mathrm{pc^{-2}},$$

used as shown above. Global masses and lengths are correspondingly written as $\log_{10}(M/M_\odot)$ and $\log_{10}(R_d/\mathrm{kpc})$; the gas fraction is already dimensionless. Before ridge fitting, all input features are standardized solely using the respective training set.

The candidate structure is not specified using the outer test galaxies. The following are selected solely within the outer training data:

1. the kernel scale λ,
2. the global feature form,
3. the ridge regularization.

The statistically tested object is therefore not a single equation set after the fact on the full data set, but a selection and prediction procedure based entirely on training data.

2.5. Fully Nested Galaxy-Separated Validation

Outer generalization is determined with five `GroupKFold` folds. All radial points of a galaxy are assigned entirely either to training or to test. Within each outer training set, a further four-fold galaxy-separated cross-validation is performed for model and regularization selection. Only then is the selected model applied to the outer test galaxies.

The RAR is tested under the same outer separation. Its reference form is

$$g_{\mathrm{RAR}}(g_{\mathrm{bar}}) = \frac{g_{\mathrm{bar}}}{1 - \exp[-\sqrt{g_{\mathrm{bar}}/g^{\dagger}}]},$$

where $g^{\dagger}$ is determined in every outer fold solely from the training galaxies. The field model and RAR are compared on exactly the same outer positive complete measurement points.

The RMSE differences are additionally evaluated with a paired galaxy-wise bootstrap. Complete galaxy blocks are sampled with replacement; a difference is considered unambiguous only when the 95% interval of the paired RMSE difference excludes zero.

2.6. Conditional OOF Galaxy-Profile Reassignment Test

For the null control, the fully independent out-of-fold predictions remain unchanged. Within each outer test fold, the observed excess profiles are randomly reassigned to other test galaxies. With differing radial grids, the donor profile is linearly interpolated onto the recipient profile in $\log_{10}(r/R_d)$. Conditional on the already trained prediction procedure, the test therefore asks whether predictions match the correct held-out galaxy profiles better than randomly reassigned profiles.

For each quality sample, 2000 permutations are calculated. The Monte Carlo value is reported with the add-one correction:

$$p_{\mathrm{MC}} = \frac{1 + N_{\mathrm{perm}}(\mathrm{RMSE}_{\mathrm{perm}} \leq \mathrm{RMSE}_{\mathrm{real}})}{2001}.$$

2.7. Galaxy-Wise Coefficient Bootstrap

To test sign stability, complete galaxies are sampled with replacement within each outer training fold. The fold-specific feature form and ridge regularization, determined solely

from training data, remain fixed; the model is refitted on every bootstrap sample. For each of the 15 outer folds, 300 replicates are calculated, yielding 4500 fits in total. Standardized coefficients, 95% intervals, and sign proportions are evaluated.

2.8. Machine-Learning Reference Models

Random Forest, Extra Trees, and HistGradientBoosting receive the same features and the same training and test galaxies as the ridge baseline model in every outer fold. Their hyperparameters are selected solely through inner galaxy-separated GroupKFold validation. A dummy mean model serves as an elementary lower baseline. Differences from the baseline model and the foldwise fitted RAR are assessed with paired galaxy-wise bootstrap intervals. The first two methods are tree-ensemble methods, and gradient boosting is used here only as a flexible predictive reference, not as a physical model [5–7].

2.9. Secondary Saturation Diagnostic

A soft nonlinear correction,

$$S_{\mathrm{soft}}(X) = \log_{10}\left(1 + 10^{\log_{10} X - \log_{10} X_{\mathrm{sat}}}\right)$$

is no longer treated as a component of the primary model. It serves as a secondary diagnostic of a possible negative high-organization curvature. In addition to the original form, an orthogonalized variant is tested, in which the component of S_{soft} that can be reconstructed linearly from the baseline features is removed solely using the training galaxies. This tests only the additional nonlinear residual component.

3. Results

3.1. Fully Nested Predictive Performance

Predictive performance is assessed from the reconstructed total acceleration. The baseline model reaches the level of the foldwise fitted RAR in all three quality samples:

Table 2: Fully nested galaxy-separated predictive performance. Δ denotes baseline model minus RAR.

Sample	Baseline model	RAR	Δ	95% interval of Δ
$Q = 1$	0.145441	0.146732	-0.001291	[-0.005837, 0.003306]
$Q = 1 + 2$	0.161875	0.164144	-0.002269	[-0.011810, 0.007525]
$Q = 1 + 2 + 3$	0.173075	0.176708	-0.003633	[-0.013297, 0.006327]

All validation results are reported in machine-readable tables and certified with SHA-256 checksums. The following figures are visual summaries of the same certified out-of-fold results and do not alter either the validation design or the reported metrics.

Figure 1 shows four representative galaxies from the primary $Q = 1 + 2$ sample. They were not selected for visual agreement, but deterministically at four percentiles of the galaxy-wise OOF error distribution. Black points show the observed rotation curves, dashed gold curves the baryonic contribution, and the blue curve the median reconstruction across four pre-specified frozen kernel variants. The pale-blue band shows their actual spread.

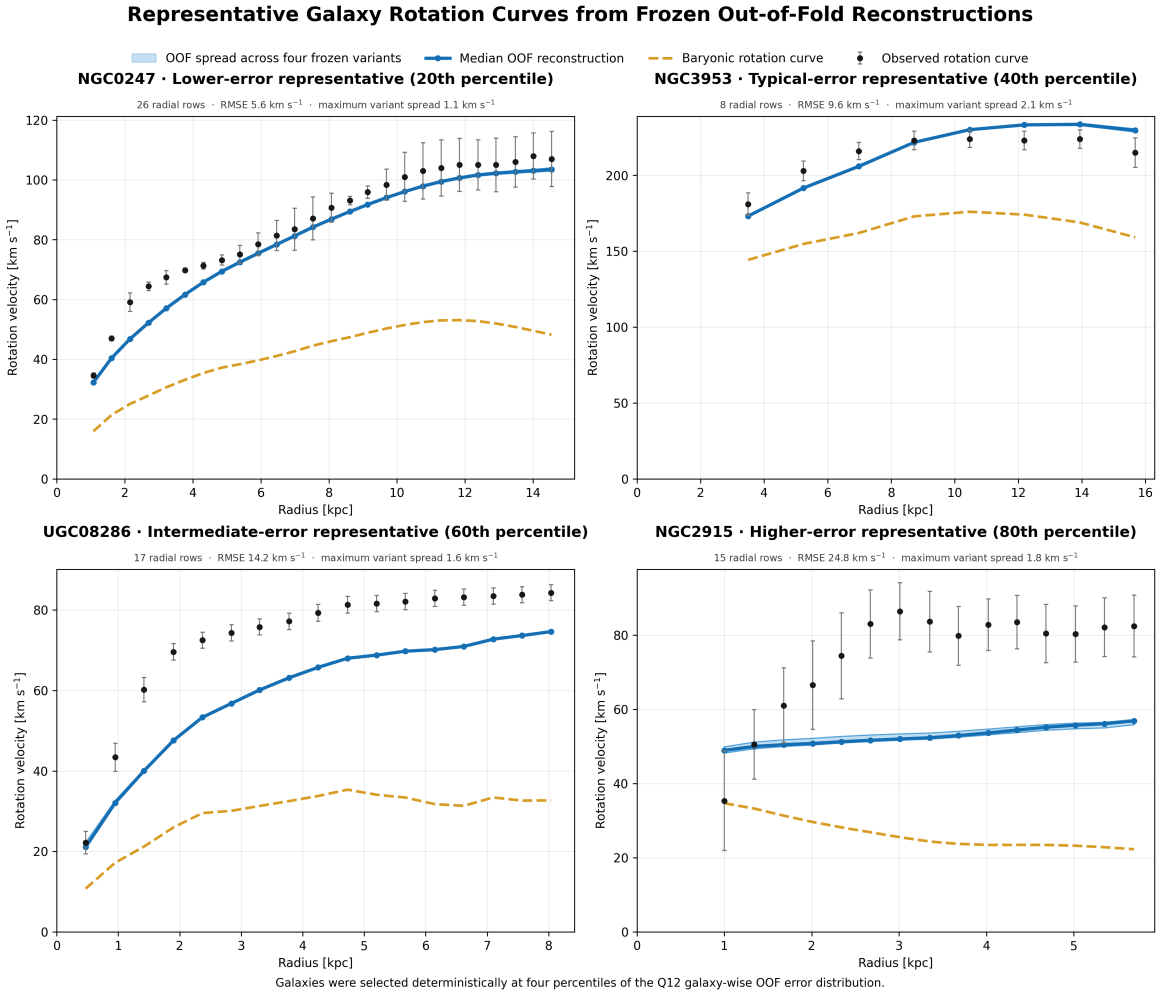


Figure 1: Representative $Q = 1 + 2$ rotation curves from frozen out-of-fold reconstructions. Black points with error bars show the observed rotation curve; the dashed gold curve shows the baryonic contribution. The solid blue curve is the median reconstruction across four pre-specified frozen kernel variants. The pale-blue band, bounded by thin blue lines, shows their actual minimum-to-maximum spread. The four galaxies were selected deterministically at the 20th, 40th, 60th, and 80th percentiles of the galaxy-wise $Q = 1+2$ OOF error distribution and were not selected for visual agreement.

The residual structure is not homogeneous across the acceleration plane. At lower accelerations, the reconstruction tends to overestimate the observed total acceleration; at higher accelerations relative to the baryonic acceleration, it tends to underestimate it. In the central region of the data distribution, the median residuals are predominantly close to zero. Figure 2 makes this regularization-related compression visible without interpreting it as a new

physical component.

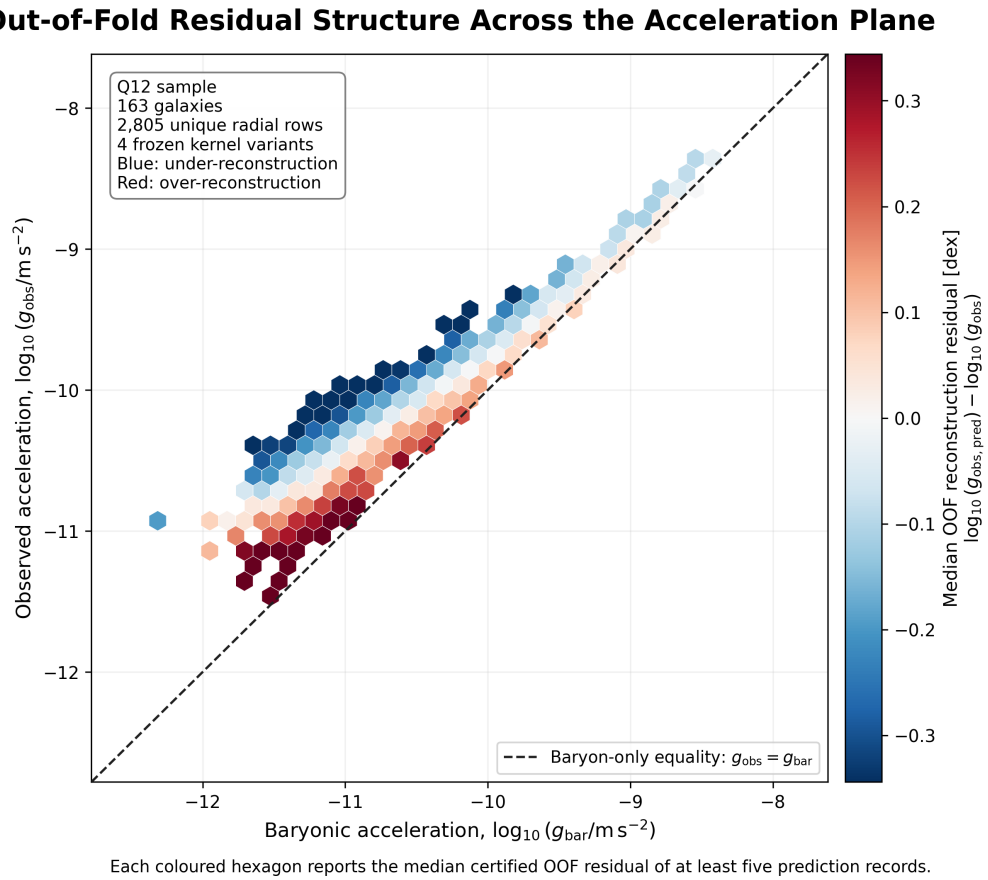


Figure 2: Out-of-fold residual structure in the $Q = 1 + 2$ acceleration plane. Each hexagon shows the median residual between the frozen OOF reconstruction and the observed total acceleration for at least five prediction records. Blue regions indicate systematic under-reconstruction, red regions systematic over-reconstruction. The dashed line marks the equality $g_{obs} = g_{bar}$, at which the baryonic component alone would explain the observed acceleration. The 11 220 prediction records correspond to 2805 unique radial points from 163 galaxies, evaluated under four pre-specified frozen kernel variants.

The nominal RMSE values of the baseline model are slightly lower in all three samples. However, all paired 95% intervals include zero. Thus neither superiority nor inferiority relative to the RAR is established. The robust conclusion is that the nonlocal field-response procedure, selected entirely from training data, and the foldwise fitted RAR attain statistically indistinguishable prediction errors.

The selected kernel scale is particularly stable in the high-quality sample: $\lambda = 0.25$ is selected in all five outer folds. For $Q = 1 + 2$, $\lambda = 0.25$ is selected in two and $\lambda = 0.5$ in three folds. For $Q = 1 + 2 + 3$, the selections are distributed across $\lambda = 0.25$, 0.5, and 0.75. The finding therefore supports a short-to-intermediate nonlocal organizational scale, but not one exactly universal value.

3.2. Sensitivity to Positive Excess Points Close to Zero

The strict filter $Z_{\rm obs} > 2$ retains 1652 of 1859 points and all 99 galaxies in $Q = 1$, 2446 of 2805 points from 160 of 163 galaxies in $Q = 1 + 2$, and 2500 of 2896 points from 167 of 175 galaxies in $Q = 1 + 2 + 3$. Thus 88.9%, 87.2%, and 86.3% of the primary positive analysis points remain.

Table 3: Predictive performance under the primary positivity filter and the strict $Z_{\rm obs} > 2$ filter. Δ denotes baseline model minus foldwise fitted RAR.

Sample	Filter	Points	Galaxies	Baseline model	RAR	Δ
$Q = 1$	$g_{\rm field} > 0$	1859	99	0.145441	0.146732	-0.001291
	$Z_{\rm obs} > 2$	1652	99	0.131489	0.134321	-0.002832
$Q = 1 + 2$	$g_{\rm field} > 0$	2805	163	0.161875	0.164144	-0.002269
	$Z_{\rm obs} > 2$	2446	160	0.136221	0.139407	-0.003186
$Q = 1 + 2 + 3$	$g_{\rm field} > 0$	2896	175	0.173075	0.176708	-0.003633
	$Z_{\rm obs} > 2$	2500	167	0.142462	0.142012	0.000450

Even under the strictest filter, the paired 95% intervals of the difference from the RAR include zero: $[-0.010048, 0.005037]$, $[-0.009312, 0.002957]$, and $[-0.006791, 0.007384]$ dex. Across all four filters, three quality samples, and five outer folds, the kernel coefficient remains positive in 60 of 60 fits. The radial term is negative in 60 of 60 fits; the local disk term is positive in 58 of 60 fits and in 45 of 45 fits at the three stricter thresholds. RAR-level accuracy and the sign structure are therefore not produced by formally positive points close to zero.

With stricter filters, the selected exact kernel scale shifts more often toward larger λ values. What is robust is therefore the existence and positive sign of a cross-radial coupling; this audit does not imply one universal physical range.

3.3. OOF Galaxy-Profile Reassignment Test

None of the 2000 randomly reassigned galaxy-profile ensembles reaches the real field RMSE:

Table 4: Conditional out-of-fold galaxy-profile reassignment test.

Sample	Real RMSE	Null mean	Best null model	$p_{\rm MC}$
$Q = 1$	0.313284	0.451863	0.366239	1/2001
$Q = 1 + 2$	0.335103	0.565395	0.449886	1/2001
$Q = 1 + 2 + 3$	0.358758	0.611101	0.489285	1/2001

The test shows that the independent predictions specifically match the correct held-out galaxy structure. It is to be interpreted as a conditional reassignment test and does not replace a full permutation test with reselection of the model under every null realization.

3.4. Coefficient Stability

The nonlocal kernel is the most stable individual structural contribution. It has a positive sign in all 4500 bootstrap fits, and its 95% intervals lie entirely above zero in all three samples:

Table 5: Galaxy-wise bootstrap of standardized core coefficients.

Sample	Coefficient	Median	95% interval	Positive proportion
$Q = 1$	Kernel	0.266522	[0.129800, 0.398589]	1.000
	Radius	-0.056084	[-0.163049, 0.050506]	0.141
	Local disk	0.084861	[-0.020804, 0.167067]	0.940
$Q = 1 + 2$	Kernel	0.267269	[0.159193, 0.408416]	1.000
	Radius	-0.093776	[-0.168897, -0.012784]	0.016
	Local disk	0.040374	[-0.024942, 0.103559]	0.887
$Q = 1 + 2 + 3$	Kernel	0.275880	[0.158111, 0.467879]	1.000
	Radius	-0.088145	[-0.205710, -0.000815]	0.025
	Local disk	0.048429	[-0.059136, 0.120075]	0.731

The radial term is stably negative in the expanded samples, whereas its interval crosses zero in $Q = 1$. The local disk term is positive in the point estimates and in most bootstrap fits, but its pooled 95% intervals include zero in all three samples. It is therefore described as a positive tendency, not as an independently established universal contribution.

3.5. Robustness to Kernel Construction

The complete kernel calculation was first checked numerically against the stored original columns. For all six candidate scales, the maximum relative deviations ranged from 2.4×10^{-16} to 3.0×10^{-16}, i.e. at machine precision. All kernel variants were then evaluated using the same outer galaxy folds and a two-stage model selection repeated exclusively within the training galaxies.

Table 6: Comparison of the main kernel and the leave-one-out kernel after full reselection. $\Delta_{\mathrm{LOO-main}}$ denotes leave-one-out minus the main kernel.

Sample	Main kernel	Leave-one-out	$\Delta_{\mathrm{LOO-main}}$	95% interval	RAR
$Q = 1$	0.145441	0.147835	0.002394	[-0.000777, 0.006264]	0.146732
$Q = 1 + 2$	0.161875	0.163324	0.001449	[-0.000303, 0.003498]	0.164144
$Q = 1 + 2 + 3$	0.173075	0.173458	0.000383	[-0.001140, 0.001900]	0.176708

All three intervals include zero. Thus, after renewed scale and feature selection, the kernel with the self-contribution fully removed is not statistically distinguishable from the main kernel. At the same time, its standardized coefficient is positive in all 15 outer-fold models. The ranges are 0.179–0.307 for $Q = 1$, 0.242–0.306 for $Q = 1+2$, and 0.231–0.355 for $Q = 1+2+3$. The positive kernel result therefore cannot be explained solely by reintroducing the local source value through $K_{ii} = 1$.

The local self-contribution is nevertheless not negligible. Its median share of the kernel

source value is approximately 21.3%, 14.8%, and 13.3% in the three quality samples. It slightly improves the numerical representation, but is not required for the positive cross-radial component. Weighting by $\Delta \ln r$ and the sign-preserving gas-source variant change prediction errors by only a few 10^{-4} dex; their paired intervals relative to the main kernel include zero. The result is therefore neither a consequence of uneven radial sampling density nor dependent on treating negative SPARC gas contributions in absolute value.

3.6. Comparison with Machine-Learning Baselines

The nested ridge baseline model achieves the lowest nominal total-acceleration RMSE in all three samples:

Table 7: Galaxy-separated comparison of the baseline model, the RAR, and generic regressors.

Sample	Baseline model	RAR	Extra Trees	Random Forest	Hist. GB	Dummy
$Q = 1$	0.145441	0.146732	0.151938	0.160460	0.172640	0.215328
$Q = 1 + 2$	0.161875	0.164144	0.166935	0.174355	0.193265	0.234835
$Q = 1 + 2 + 3$	0.173075	0.176708	0.175782	0.178651	0.189047	0.243635

Random Forest is clearly weaker than the baseline model in $Q = 1$ and $Q = 1 + 2$; HistGradientBoosting is clearly weaker in all three samples. Extra Trees is nominally above the baseline model, but the paired 95% intervals of the difference include zero in every sample. The appropriate conclusion is therefore not that the linear baseline model significantly outperforms every flexible regressor. The robust result is that none of the tested machine-learning baselines demonstrably improves upon the interpretable baseline model.

3.7. Secondary Saturation Diagnostic

The additional saturation form provides no consistent incremental predictive gain:

Table 8: Baseline model and secondary saturation diagnostic; total-acceleration RMSE is reported.

Sample	Baseline model	Original saturation	Orthogonalized saturation	RAR
$Q = 1$	0.145441	0.145240	0.146431	0.146732
$Q = 1 + 2$	0.161875	0.164905	0.163746	0.164144
$Q = 1 + 2 + 3$	0.173075	0.172515	0.171745	0.176708

All paired galaxy-wise bootstrap intervals of the saturation difference relative to the baseline model include zero. In addition, the original saturation variable is strongly collinear with the baseline features; the median diagnostic VIF values are approximately 40.6, 269.2, and 367.8. After the linearly redundant component is removed, the nonlinear residual correction remains negative in all 15 outer folds. This sign is a reproducible secondary observation, but not yet evidence for a universal saturation law, a uniquely determined transition scale, or additional predictive gain.

3.8. Compact C3 Candidate, Model Reduction, and Freeze for an External Test

The present SPARC analyses are the development and internal-validation basis for the empirical response examined here; they do not constitute external confirmation or a derivation of a fundamental mechanism. For clarity, *M19* denotes the full nonlocal reference response used in the component-ablation and joint out-of-fold (OOF) analyses. M19 is an internal model label, not a citation to an external publication.

A supplementary group-separated audit of 175 galaxies and 3389 valid radial points examined which components of this response remain necessary when predicting previously held-out galaxies. The 3389-point audit uses the full set of valid radial rows before the positivity and complete-feature restrictions applied to the logarithmic baseline model; it is therefore distinct from the 1859, 2805, and 2896-point primary analysis samples defined above.

The radial gradient of the field response is the clearest individual result in this audit. Removing it significantly worsens galaxy-wise OOF prediction relative to the M19 reference (mean ΔRMSE = +0.00814 dex; $p = 0.010$). The enforced linear scale form $q = 1$ is also worse (ΔRMSE = +0.00479 dex; $p = 0.028$). By contrast, the precise saturation form and the precise choice among compact nonlinear scale forms are not uniquely determined by the present data. Within this model family, the data thus support a spatially structured, non-purely-local field response, but not yet its unique microphysical form.

The exact M19 response form used in the audit is `M19_full_free_q_saturated_gradient`:

$$g_{\rm cum}(r) = \frac{2}{r^2}\int_0^r r'\, g_{\rm bar}(r')\,{\rm d}r', \tag{1}$$

$$\omega(r) = g_{\rm bar}(r)^{1-\eta}\, g_{\rm cum}(r)^{\eta}, \tag{2}$$

$$s_q(r) = g_{\rm ref}\left(\frac{\omega(r)}{g_{\rm ref}}\right)^{q}, \tag{3}$$

$$\Delta g_{\rm M19}(r) = A\, g_{\rm max}\, \tanh\!\left(\frac{s_q(r)}{g_{\rm max}}\right), \tag{4}$$

$$g^{\rm M19}_{\rm pred}(r) = g_{\rm bar}(r) + \Delta g_{\rm M19}(r) + \lambda \frac{\rm d}{{\rm d}r}\left[r\,\Delta g_{\rm M19}(r)\right]. \tag{5}$$

M19 therefore has six fitted parameters: A, $g_{\rm ref}$, $g_{\rm max}$, η, q, and λ. It contains a nonlocal source quantity, a freely fitted scale exponent q, a saturation term, and a radial-gradient term.

For a future external test, a deliberately simpler, algebraically identifiable representation was specified:

$$g_{\rm cum}(r) = \frac{2}{r^2}\int_0^r r'\, g_{\rm bar}(r')\,{\rm d}r', \tag{6}$$

$$\omega(r) = g_{\rm bar}(r)^{1-\eta}\, g_{\rm cum}(r)^{\eta}, \tag{7}$$

$$\Delta g(r) = K\sqrt{\omega(r)}, \tag{8}$$

$$g^{\rm C3}_{\rm pred}(r) = g_{\rm bar}(r) + \Delta g(r) + \lambda \frac{\rm d}{{\rm d}r}\left[r\,\Delta g(r)\right]. \tag{9}$$

This form is hereafter denoted *C3*. It is the candidate with fixed $q = 0.5$, a linear unsaturated

response, and a retained radial-gradient term within the M19 model family. The algebraic equality test to machine precision applies to the two parameterizations of this C3 form; it does not assert identity between C3 and the full M19 reference model with free q and saturation. The two linear C3 parameterizations are exactly related by $K = A\sqrt{g_{\mathrm{ref}}}$.

The direct joint OOF comparison with the full M19 reference model in Equation 5 yields no statistically robust overall difference: for C3 relative to M19, the mean galaxy-wise difference is $\Delta\mathrm{RMSE} = +0.00114$ dex (two-sided Wilcoxon test $p = 0.402$; bootstrap 95 % interval $[-0.00249, +0.00480]$ dex). Several small individual differences must therefore not be added into an overall gain. C3 was selected for parsimony and algebraic identifiability, not because it is unambiguously superior to M19.

The parameters were then estimated once on the full SPARC set and fixed for a future external evaluation:

$$\log_{10} K = -5.306022, \qquad K = 4.942856 \times 10^{-6}, \qquad \eta = 0.925228, \qquad \lambda = 0.415409. \quad (10)$$

The corresponding descriptive fit to the same SPARC set gives RMSE = 0.192017 dex, MAE = 0.134073 dex, and $R^2 = 0.884091$. These metrics are explicitly not external validation.

Every subsequent independent evaluation is subject to a lock rule: the equation, parameters, and preprocessing must not be readjusted. An eligible catalogue must be independent of SPARC and provide positive radii together with documented baryonic and observed accelerations for each galaxy. The primary reporting comprises g_{pred} versus g_{obs}, the complete residual accounting, and galaxy-wise error measures. As a second representation, the predicted excess $\Delta g + \lambda\, \mathrm{d}(r\Delta g)/\mathrm{d}r$ is compared with the observed excess $g_{\mathrm{obs}} - g_{\mathrm{bar}}$; non-positive observed excesses are reported in full and not removed in a result-dependent manner.

3.9. Spectral and Population Context

The regimes and parameters derived from the rotation curves were additionally compared with publicly available broadband quantities. This step does not retest the C3 equation; it addresses only the narrower question of whether the structure visible in the rotation curves can already be replaced by a simple global population or colour quantity.

An unambiguous match to the official WISE table was established for 111 of the 175 SPARC galaxies [10]. In this subsample, 3.6-μm luminosity alone explains WISE colour only to a limited degree ($R^2 = 0.345$); adding H I mass and disk radius increases the value to $R^2 = 0.394$, and further structural quantities to $R^2 = 0.417$. Broadband colour thus contains physically meaningful but insufficient information about galactic structure.

Crucially, the tested WISE, GALEX, and SDSS broadband quantities do not yield a robust separation of the q regimes determined from the rotation curves. Nor does the comparison of population colour with the remainder of the nonlocal baseline model show a robust linear coupling: in the WISE subsample, the galaxy-level result is $r = -0.142$ with $p \approx 0.134$, and in the high-quality $Q = 1$ subsample it is $r = -0.042$ with $p \approx 0.726$. Colour can therefore replace neither the nonlocal kernel nor the radial gradient.

A narrower, explicitly exploratory observation nevertheless remains. Across all radial points,

a smoothed WISE-based M/L signal shows a negative association with the model residual (111 galaxies: Pearson $r = -0.221$, Spearman $\rho = -0.332$, $p = 0.019$); it is stronger in $Q = 1$ (71 galaxies: Pearson $r = -0.333$, Spearman $\rho = -0.379$, $p = 0.0039$). Because this analysis was performed after development of the core model and WISE coverage is incomplete, it is not claimed as additional model gain or as spectral field coupling. It only motivates a prespecified independent test with fully propagated photometric and mass-model uncertainties.

Smoothed reference spectra also preserve the expected broad sequence from elliptical/S0 systems through spirals to starbursts. This shows that a broadband smoothed description does not completely erase real population differences. It does not, however, demonstrate that this spectral ordering causes the nonlocal dynamics or determines the C3 parameters.

3.10. Synthesis of the Empirical Results

The fully corrected analysis supports three tiered conclusions. First, the positive acceleration excess in entirely held-out galaxies can be predicted from a nonlocal, baryon-anchored, scale-dependent model family. Second, this procedure attains prediction accuracy that is statistically indistinguishable from the foldwise fitted RAR, while no tested generic machine-learning baseline provides a demonstrable improvement. Third, the positive contribution of the nonlocal kernel is exceptionally stable. It persists under stricter positivity filters and remains positive in all outer folds even when the local self-contribution is completely removed. Radial and local disk contributions are more differentiated: the radial term is stably negative in the expanded samples, whereas the local disk term remains a predominantly positive but not unambiguously non-zero tendency.

A negative nonlinear high-organization correction occurs repeatedly. Its precise form, transition scale, and incremental predictive performance are not uniquely determined by the present parameterization. It is therefore carried into the theoretical discussion as a secondary hypothesis, not as a primary empirical conclusion.

4. Further Consistency Tests

The following tests examine whether the primary nonlocal field-response model also remains consistent in related representations of the rotation-curve data. MDAR, BTFR, and inner mass-slope structures are not fully independent data sets because they are derived from the same or closely related SPARC quantities. They are nevertheless necessary consistency checks: a viable field-response model must not only reconstruct the pointwise acceleration g_{obs}, but must also remain compatible with the acceleration ratio, the global baryonic scaling relation, and the inner structure of the rotation curves.

These tests therefore do not extend the data basis; instead, they tighten the internal requirement imposed on the model. A mere point-by-point fit would be insufficient if it produced contradictory structure in MDAR space, BTFR residuals, or inner rotation-curve organization.

4.1. MDAR Consistency Test

In addition to direct prediction of $g_{\rm obs}$, the primary nonlocal field-response model was tested in MDAR/RAR space [2, 3]. The dynamical acceleration ratio considered was

$$D_{\rm MDAR} = \frac{g_{\rm obs}}{g_{\rm bar}},$$

as well as the corresponding model and RAR quantities

$$D_{\rm field} = \frac{g_{\rm obs,field}}{g_{\rm bar}}, \qquad D_{\rm RAR} = \frac{g_{\rm RAR}}{g_{\rm bar}}.$$

The evaluation was performed in logarithmic $\log_{10} D_{\rm MDAR}$ space. Because this representation is mathematically closely linked to direct acceleration prediction, the MDAR test does not provide an independent external data set. It does, however, tighten the internal consistency requirement: the model must not only reconstruct $g_{\rm obs}$, but also preserve the mass-discrepancy representation $g_{\rm obs}/g_{\rm bar}$ without contradictory structure. This is indeed the case. In the high-quality $Q = 1$ sample, the baseline model is also effectively at the level of the empirical RAR in MDAR space. In the expanded $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples, the modest improvement relative to the RAR found in the direct acceleration analysis remains.

The MDAR test thus confirms the previously identified structure in a different representation: the nonlocal field response not only reproduces the observed total acceleration, but also remains consistent with the dynamical ratio $g_{\rm obs}/g_{\rm bar}$.

4.2. BTFR Consistency Test

In addition to pointwise acceleration structure, the baryonic Tully–Fisher relation (BTFR) was examined as a galaxy-wide consistency relation [8]. While RAR and MDAR compare radial acceleration points, the BTFR describes a global relation between baryonic mass and asymptotic rotation velocity:

$$\log_{10} M_{\rm bar} = A + B \log_{10} V_{\rm flat}.$$

For the high-quality $Q = 1$ sample, the result was

$$A = 2.585720, \qquad B = 3.557839,$$

with

$$\mathrm{RMSE}_{\rm BTFR} = 0.226093 \text{ dex}, \qquad R^2 = 0.924208.$$

For the expanded $Q = 1 + 2$ sample, the result was

$$A = 2.916817, \qquad B = 3.413261,$$

with

$$\mathrm{RMSE}_{\mathrm{BTFR}} = 0.247747 \text{ dex}, \qquad R^2 = 0.910985.$$

For the full $Q = 1 + 2 + 3$ sample, the result was

$$A = 3.399839, \qquad B = 3.189205,$$

with

$$\mathrm{RMSE}_{\mathrm{BTFR}} = 0.288828 \text{ dex}, \qquad R^2 = 0.881624.$$

The increasing scatter and shallower slope when lower quality classes are included are consistent with the expectation that the high-quality $Q = 1$ sample provides the most stable global scaling relation.

Of particular interest is the relation between BTFR residuals and the galaxy-wise residuals of the primary field-response model. BTFR residuals were compared with median field-response-model residuals at galaxy level. The analysis was performed consistently within each sample, so that every galaxy was included only once in the respective quality sample.

For $Q = 1$, the relation between the BTFR residual and the median field-response residual was

$$r_{\mathrm{Spearman}} = -0.644037, \qquad p = 1.70 \times 10^{-11}.$$

For $Q = 1 + 2$, it was

$$r_{\mathrm{Spearman}} = -0.706910, \qquad p = 7.88 \times 10^{-21}.$$

For $Q = 1 + 2 + 3$, it was

$$r_{\mathrm{Spearman}} = -0.703741, \qquad p = 1.75 \times 10^{-21}.$$

The BTFR scatter is therefore systematically coupled to the field-response residual structure. Galaxies that deviate relative to the global BTFR simultaneously show systematic deviations in radial field-excess modelling. In this analysis, global mass–velocity scaling and radial excess structure are not independent phenomena, but two coupled representations of the same galactic dynamics.

This result is an internal consistency check. It does not replace an independent data set, but tightens the requirement on the model: a viable field-response description must not only be pointwise compatible with RAR or MDAR, but must also remain compatible with the global baryonic scaling relation. This coupling between global BTFR structure and radial field-response residual structure is observed in all three quality selections.

5. Discussion

The main result is a reproducible predictive structure, not a unique physical interpretation. Under nested galaxy-separated validation, the nonlocal model reaches the same empirical accuracy class as a foldwise refitted RAR while retaining a stable positive cross-radial kernel. The conditional null test based on profile reassignment shows that this result depends on the observed correspondence between a galaxy's radial baryonic profile and its acceleration-excess profile, rather than solely on the distribution of individual data points.

The analysis has important limitations. It uses the SPARC development sample; although all tuning is separated from the held-out galaxies, the candidate family was developed on this data set. The reported uncertainty analysis is not a complete joint propagation of the uncertainties in distance, inclination, mass-to-light ratio, and baryonic decomposition. The C3 form is therefore a frozen candidate for an external test, not a uniquely identified law. MDAR and BTFR provide internal consistency checks only.

The next decisive test is a preregistered external validation using independent resolved rotation curves and mass models, without parameter adjustment and with complete uncertainty treatment. Failure of the fixed C3 form, or failure to obtain a positive nonlocal contribution under the same predefined procedure, would directly constrain this empirical model family.

6. Conclusion

This work shows that the positive acceleration excess in SPARC rotation curves can be reconstructed reproducibly by a fully nested, galaxy-separated, nonlocal empirical procedure. For $Q = 1$, $Q = 1 + 2$, and $Q = 1 + 2 + 3$, the primary baseline model attains total-acceleration RMSE values of 0.145441, 0.161875, and 0.173075 dex. The foldwise fitted RAR attains 0.146732, 0.164144, and 0.176708 dex for the same measurements. The paired galaxy-wise intervals include zero in every case; the two procedures therefore belong to the same empirical accuracy class.

The strongest structural finding is the stable positive contribution of the nonlocal dynamical kernel. It remains positive in all 4,500 bootstrap fits and is clearly separated from zero in every quality sample. For the leave-one-out kernel with $K_{ii} = 0$, the coefficient likewise remains positive in all 15 outer folds; its predictive performance after full reselection is not statistically distinguishable from that of the main kernel. Stricter positivity filters up to $Z_{\mathrm{obs}} > 2$ preserve the same sign structure and RAR-level accuracy. The radial term is stably negative in the expanded samples. The local disk term shows a predominantly positive but not conclusively established tendency.

The conditional OOF galaxy-profile reassignment test yields $p_{\mathrm{MC}} = 1/2001$ in all three samples; no randomly reassigned profile ensemble attains the real model performance. Random Forest and HistGradientBoosting are generally weaker, whereas Extra Trees remains statistically close to the baseline model. None of the tested machine-learning baselines demonstrably improves the interpretable model family.

The additional saturation correction, also tested conditionally by regime, provides neither a consistent incremental predictive gain nor a uniquely identified transition scale. Saturation

therefore remains a secondary, empirically motivated hypothesis and is not treated as an established primary result.

As a provisional, parsimonious representation for the next independent rotation-curve test, C3 with a fixed nonlocal $q = 0.5$ response and a radial gradient was frozen. C3 is algebraically equivalent to its earlier fixed-$q = 0.5$ notation and does not differ significantly from the full free-q, saturated M19 reference model in the internal SPARC OOF test. Its selection is therefore a transparent simplification, not a claim of a uniquely final field equation. Only an external application without parameter adjustment can confirm or falsify this candidate.

The result establishes neither a fundamental mechanism nor an alternative to particle dark matter. It defines a reproducible empirical target for future work: the robust nonlocal, baryon-anchored, scale-dependent structure, including the positive contribution of a coupling quantity constructed exclusively from other radii. The immediate next step is a preregistered external application of the frozen C3 candidate to independent rotation curves and mass models with fully propagated observational and baryonic-model uncertainties [4, 9].

Author Contributions

Conceptualization, U.Q.R.; methodology, U.Q.R.; software, U.Q.R.; validation, U.Q.R.; formal analysis, U.Q.R.; investigation, U.Q.R.; resources, U.Q.R.; data curation, U.Q.R.; writing—original draft preparation, U.Q.R.; writing—review and editing, U.Q.R.; project administration, U.Q.R. The author has read and approved the manuscript.

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Data Availability Statement

The underlying SPARC rotation-curve and mass-model data are publicly available through the original SPARC data publication cited in this manuscript. The frozen inputs, one-cell reproduction code, certified out-of-fold validation outputs, figure source data, validation certificate, and integrity checksums used in this study are publicly available in the following repository:

https://github.com/ukshinrexhepi-cloud/SPARC-Frozen-Validation-Reproduction

The exact archival snapshot associated with this manuscript is permanently available through Zenodo:

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The Zenodo archive preserves the certified repository version, including the frozen validation workflow, 60 outer-fold fits, 30,240 reconstructed out-of-fold records, the validation certificate, and the two figures with their corresponding source data.

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Conflicts of Interest

The author declares no conflict of interest.

Abbreviations

BTFR	Baryonic Tully–Fisher relation
CV	Cross-validation
MDAR	Mass-discrepancy–acceleration relation
M19	Full free-q, saturated nonlocal gradient reference model
C3	Fixed-$q = 0.5$, linear-gradient candidate model
OOF	Out-of-fold
RAR	Radial acceleration relation
RMSE	Root mean squared error
SPARC	Spitzer Photometry and Accurate Rotation Curves

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